Commute Concierge

Feature roadmap — editable, step-by-step commute planning for low-confidence transit users, built for HackUCD 2026 (Sustainability / Travel & carpooling track).

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1. Current Features, Prioritized

What's already built in the prototype, ranked by how much each one drives the judging rubric (Users first, Craft, Stay curious, AI usage) versus how replaceable it is if time runs short.

Priority	Feature	Why it matters
P0 – Built	**Editable per-leg itinerary**	The core differentiator from Google Maps/Transit – every leg (walk, bus, Luas, cycle) can be swapped independently instead of accepting one auto-routed plan.
P0 – Built	**Preference toggle (fastest / fewest transfers / cheapest)**	Directly requested by the team; re-scores every leg and recomputes totals live, giving users a sense of control without needing to understand the underlying data.
P0 – Built	**AI break-time concierge**	The required OpenRouter feature with real reasoning (not a wrapped call): given a transfer point and a free-time window, it recommends what to do and when to resume – this is the single feature judges will remember.
P0 – Built	**Automatic fallback at each leg**	Next-departure times are always visible per leg, so a missed connection never leaves the user with no information – addresses 'Craft: thoughtful error handling'.
P1 – Built	**Live disruption alerts (WebSocket)**	Server polls a GTFS-Realtime-shaped feed and pushes delay/cancellation events to connected clients in real time; currently simulated, structured to swap in the real NTA feed.
P1 – Built	**Freeform AI chat panel**	Lets users ask ad-hoc questions ('what if I miss this connection?') with the live itinerary injected as context, reinforcing AI usage beyond the break feature.
P1 – Built	**Clean, high-contrast UI**	Large text, big tap targets, plain-language labels – built specifically for the stated target user (newcomers/low-confidence commuters), not a generic dashboard look.
P2 – Built	**Stripe sandbox subscription (Pro tier)**	Satisfies the optional 5-point Stripe bonus: real Checkout Session, test payment, and the purchase actually unlocks a feature (richer AI recommendations) rather than just a button.

2. Suggested Features, Prioritized

Ranked by expected judging impact per hour of build time remaining. P1 items are the next things worth building; P3 is a 'only if everything else is done' list.

Priority	Feature	Why it matters
P1 – Next	**Real NTA GTFS-Realtime integration**	Replaces the simulated live feed with an actual pull from NTA's developer API; strengthens the 'use suitable data' judging criterion with a real, not synthetic, live signal.
P1 – Next	**First-time micro-walkthrough**	A 3-step overlay the first time the app loads ('tap here to change how you travel', 'tap here to add a break') – directly serves the 'regardless of technological aptitude' promise, which the live app doesn't yet demonstrate on its own.
P1 – Next	**Accessibility / step-free flag per option**	Lets a user filter for step-free routes; cheap to add to the existing data model and broadens 'Users first' beyond one persona.
P1 – Next	**One-tap 'explain this route simply' button**	Sends the current itinerary to the AI concierge and returns a 2-sentence plain-language summary – a second, very cheap OpenRouter use case that reinforces the AI criterion.
P2 – Later	**Save / share itinerary link**	A shareable URL (or QR code) for a planned trip – useful for the 'how would you reach these users' story (e.g. international-student orientation sharing a link).
P2 – Later	**Saved favourites / recent journeys**	Removes re-entry friction for repeat commutes; needs lightweight persistence (localStorage is enough for a demo).
P2 – Later	**Second corridor / arbitrary origin-destination**	Demonstrates the approach generalises beyond one hardcoded route; only worth it once the single-corridor demo is fully polished.
P2 – Later	**Calendar-aware arrival time**	Reads 'arrive by' from the user's next calendar event instead of manual entry – nice craft touch, not required for the core story.
P3 – Stretch	**Carpool matching for the same corridor**	Pulled from the original CSO-commute-flow idea; a bigger scope item, only sensible as a stretch goal with time to spare.
P3 – Stretch	**Trusted-contact live ETA sharing**	Share a live link of your progress with a family member/friend – strong safety story for newcomers, but adds real-time infra beyond what's needed to win the rubric.
P3 – Stretch	**Gamification for sustainable choices**	Badges/streaks for choosing lower-CO2 legs; fun but not aligned with the 'low-tech-aptitude, needs confidence' persona – validate demand before building.

3. Mobile UI Recommendations

Commuters will use this standing up, one-handed, often with a bus arriving – mobile isn't a breakpoint afterthought here, it's close to the primary use case.

High priority

- Single-column layout below 480px with a sticky top bar showing only the next action (current leg + next departure countdown), so the most time-critical info never scrolls out of view.

- Minimum 48×48px tap targets on every control, especially the mode-swap dropdown and break-duration stepper – current desktop sizing is too small for thumbs.
- Collapse the vertical leg timeline into one expandable card at a time on mobile (accordion), rather than showing all legs at once, to reduce scrolling and cognitive load for low-tech-aptitude users.
- Move the chat panel behind a floating action button (FAB) instead of an always-visible sticky panel – on a small screen the current layout would push the itinerary almost fully off-screen.

Medium priority

- Installable PWA (manifest + service worker) so the static timetable and last-fetched itinerary work with no signal underground or on a train.
- Large, high-contrast 'next departure' countdown as a persistent bottom bar, updating live, visible even while the chat or break panel is open.
- Voice input on the chat box (Web Speech API) – useful both for accessibility and for 'can't look at the phone while walking' moments.
- Haptic feedback (navigator.vibrate) when a live disruption alert arrives, since a visual-only banner is easy to miss on a packed bus.

Lower priority

- Dark mode following system preference.
- Swipe gestures between legs instead of scrolling.
- Native share-sheet integration for the 'share itinerary' feature once it exists.

4. Other Things Worth Covering

Testing the core assumption

- Rubric requires evidence, not just a built feature: run a 5-minute think-aloud test with 2–3 newcomers/international students on campus before judging – does step-by-step editability reduce their anxiety, or is it overwhelming?
- Test boundary and failure inputs explicitly: last trip of the day, a transfer with zero remaining departures, and a break longer than the gap until the next service – show what the app does in each case, not just the happy path.

Data & honesty about sources

- State clearly in the demo that route/timetable data is drawn from NTA's published schedules for one corridor, and that live disruptions are currently simulated on the same polling cadence a real GTFS-Realtime integration would use – the rubric rewards stating assumptions, not hiding them.
- If cost/CO2 figures are shown, state the SEAI conversion factors and fare assumptions used, per the 'for any claimed savings, explain what you are comparing against' requirement.

Accessibility & privacy

- Keep color contrast and focus states (already in the current CSS) – verify with a quick screen-reader pass on the itinerary cards and chat panel before judging.
- No real personal data is collected in the current prototype (no accounts, no stored journeys) – worth saying explicitly, since newcomer/anxious-commuter users will care about this.

Demo & pitch framing

- Open the pitch with the specific user and their current workaround (leaving an hour early, asking a friend), not with the feature list – matches how 'Users first' is scored.
- Narrate the AI concierge live during the demo (ask it a real question in front of judges) rather than describing it – 'demonstrated output quality and understanding of its limitations' is explicitly what's scored.
- Have one rehearsed failure-path demo (missed connection, cancelled bus) ready – it's a stronger craft signal than a second happy-path route.

Generated for the HackUCD 2026 prototype in /commute-planner.